

## *Ethernet LAN switching (PART 1)*



## OSI Model – Physical Layer

|   |              |
|---|--------------|
| 7 | Application  |
| 6 | Presentation |
| 5 | Session      |
| 4 | Transport    |
| 3 | Network      |
| 2 | Data Link    |
| 1 | Physical     |

- Defines physical characteristics of the medium used to transfer data between devices.
- For example, voltage levels, maximum transmission distances, physical connectors, cable specifications, etc.
- Digital bits are converted into electrical (for wired connections) or radio (for wireless connections) signals.
- All of the information in Day 2's video (cables, pin layouts, etc.) is related to the Physical Layer.





## OSI Model – Data Link Layer

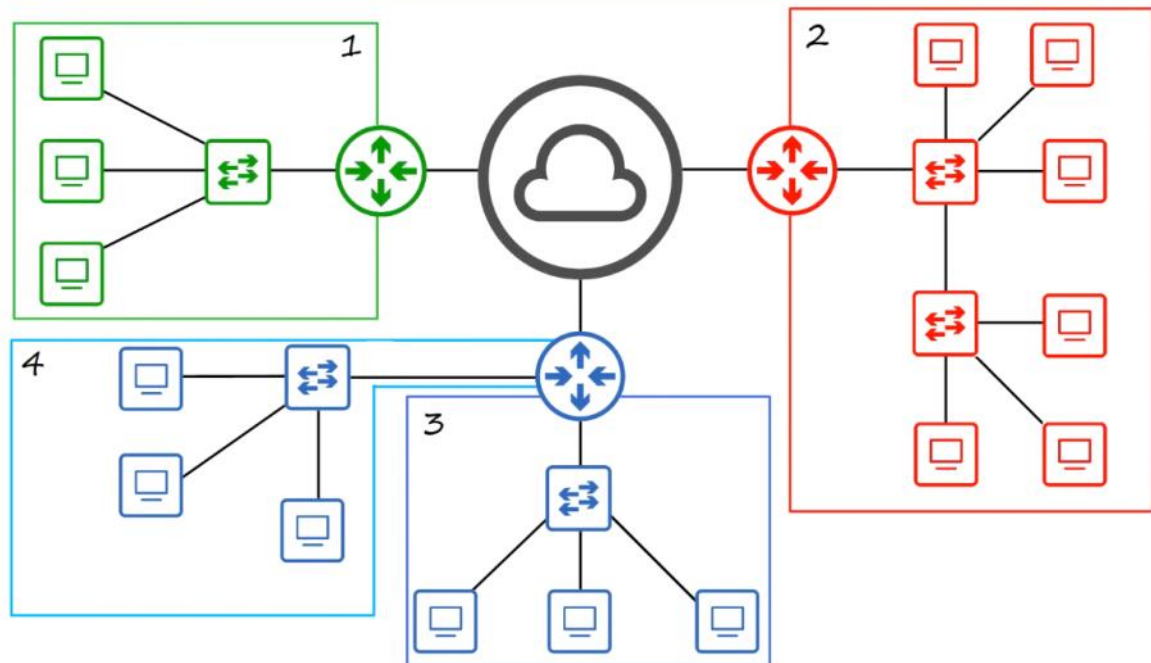
|   |                  |
|---|------------------|
| 7 | Application      |
| 6 | Presentation     |
| 5 | Session          |
| 4 | Transport        |
| 3 | Network          |
| 2 | <b>Data Link</b> |
| 1 | Physical         |

- Provides node-to-node connectivity and data transfer (for example, PC to switch, switch to router, router to router).
- Defines how data is formatted for transmission over a physical medium (for example, copper UTP cables)
- Detects and (possibly) corrects Physical Layer errors.
- Uses Layer 2 addressing, separate from Layer 3 addressing.
- Switches operate at Layer 2.





## Local Area Networks (LANs)



A router has **multiple interfaces** (ports).

Each interface connects to a **different network**.

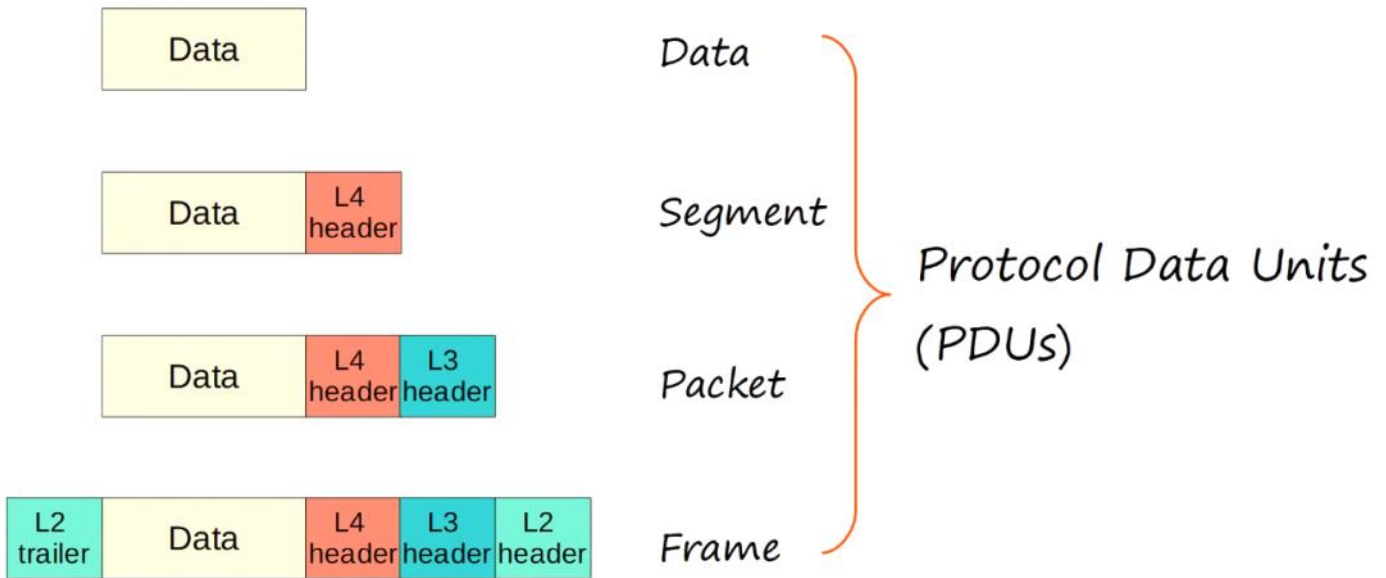
- Interface 1
- Interface 2
- Interface 3
- Interface 4

Each interface:

- Has its **own IP address**.
- Belongs to a **different subnet/network**.
- Acts as the **gateway** for that LAN.

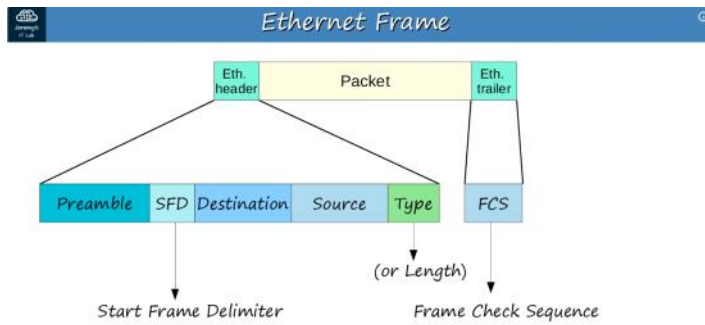


# OSI Model – PDUs



# Ethernet Frame

18 February 2026 06:04



## Preamble

- Length: 7 bytes (56 bits)
- Alternating 1's and 0's
- 10101010 \* 7
- Allows devices to synchronize their receiver clocks

## SFD

- 'Start Frame Delimiter'
- Length: 1 byte (8 bits)
- 10101011
- Marks the end of the preamble, and the beginning of the rest of the frame

## Destination & Source

### Destination

- Indicate the devices sending and receiving the frame
- Consist of the destination and source 'MAC address'
- MAC = Media Access Control
- = 6 byte (48-bit) address of the physical device

### Source

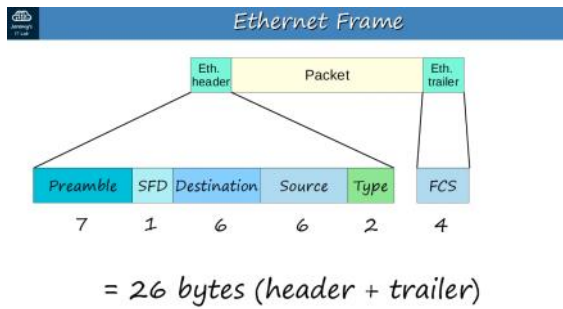
## Type / Length

### Type

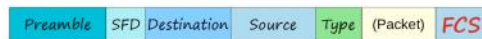
- 2 byte (16-bit) field
- A value of **1500 or less** in this field indicates the LENGTH of the encapsulated packet (in bytes)
- OR
- A value of **1536 or greater** in this field indicates the TYPE of the encapsulated packet (usually IPv4 or IPv6), and the length is determined via other methods

### Length

IPv4 = 0x0800 (hexadecimal)    IPv6 = 0x86DD (hexadecimal)  
(2048 in decimal)    (34525 in decimal)



## Frame Check Sequence (FCS)



- 'Frame Check Sequence'
- 4 bytes (32 bits) in length
- Detects corrupted data by running a 'CRC' algorithm over the received data
- CRC = 'Cyclic Redundancy Check'



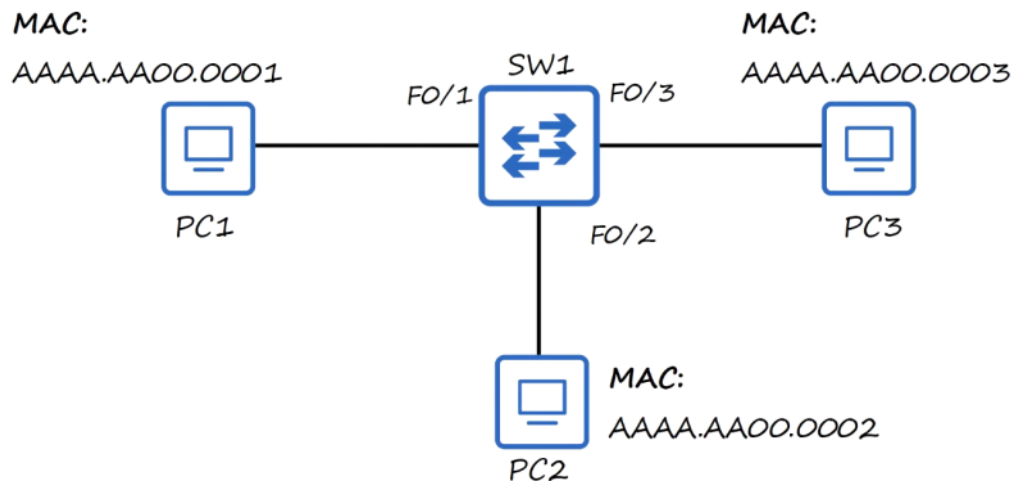
## MAC Address

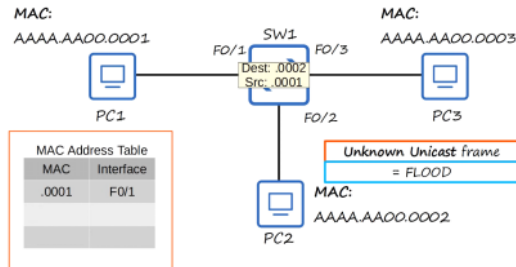
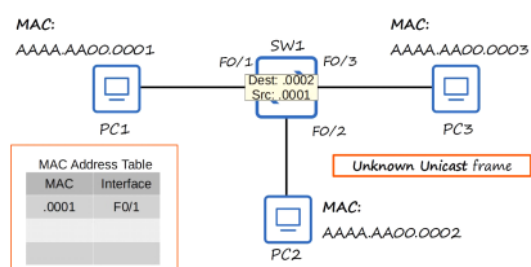
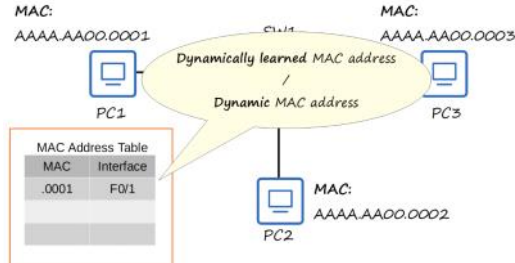
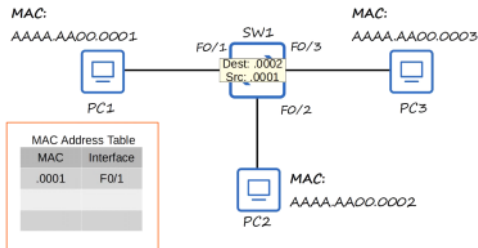
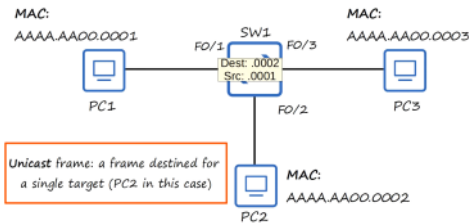
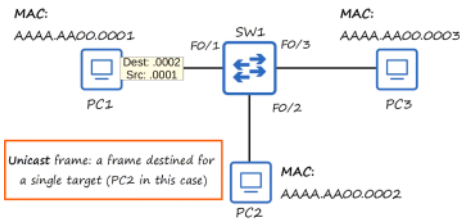
- 6-byte (48-bit) physical address assigned to the device when it is made
- A.K.A. 'Burned-In Address' (BIA)
- Is globally unique
- The first 3 bytes are the OUI (Organizationally Unique Identifier), which is assigned to the company making the device
- The last 3 bytes are unique to the device itself
- Written as 12 hexadecimal characters

## Hexadecimal

| DEC. | HEX. | DEC. | HEX. | DEC. | HEX. | DEC. | HEX. |
|------|------|------|------|------|------|------|------|
| 0    | 0    | 8    | 8    | 16   | 10   | 24   | 18   |
| 1    | 1    | 9    | 9    | 17   | 11   | 25   | 19   |
| 2    | 2    | 10   | A    | 18   | 12   | 26   | 1A   |
| 3    | 3    | 11   | B    | 19   | 13   | 27   | 1B   |
| 4    | 4    | 12   | C    | 20   | 14   | 28   | 1C   |
| 5    | 5    | 13   | D    | 21   | 15   | 29   | 1D   |
| 6    | 6    | 14   | E    | 22   | 16   | 30   | 1E   |
| 7    | 7    | 15   | F    | 23   | 17   | 31   | 1F   |

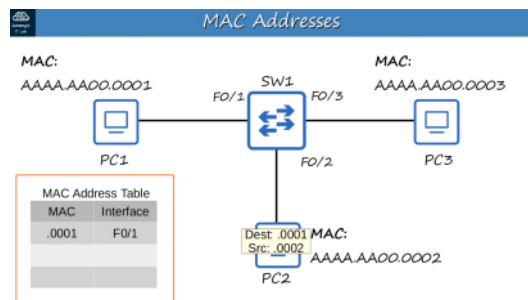
00-0a-83-b1-c0-8e



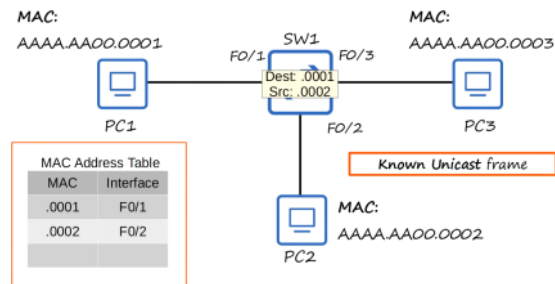


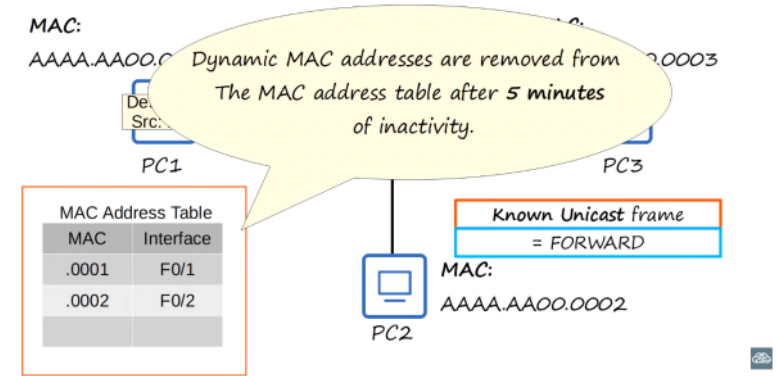
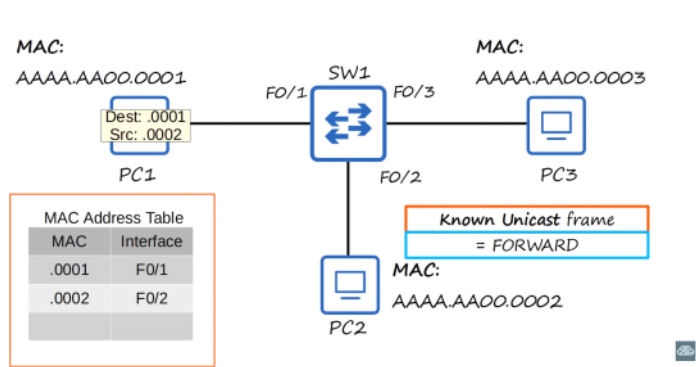
An unknown unicast frame is an Ethernet frame where:

- The destination MAC address is a specific device (not broadcast/multicast)
- But the switch does NOT know which port that MAC address belongs to



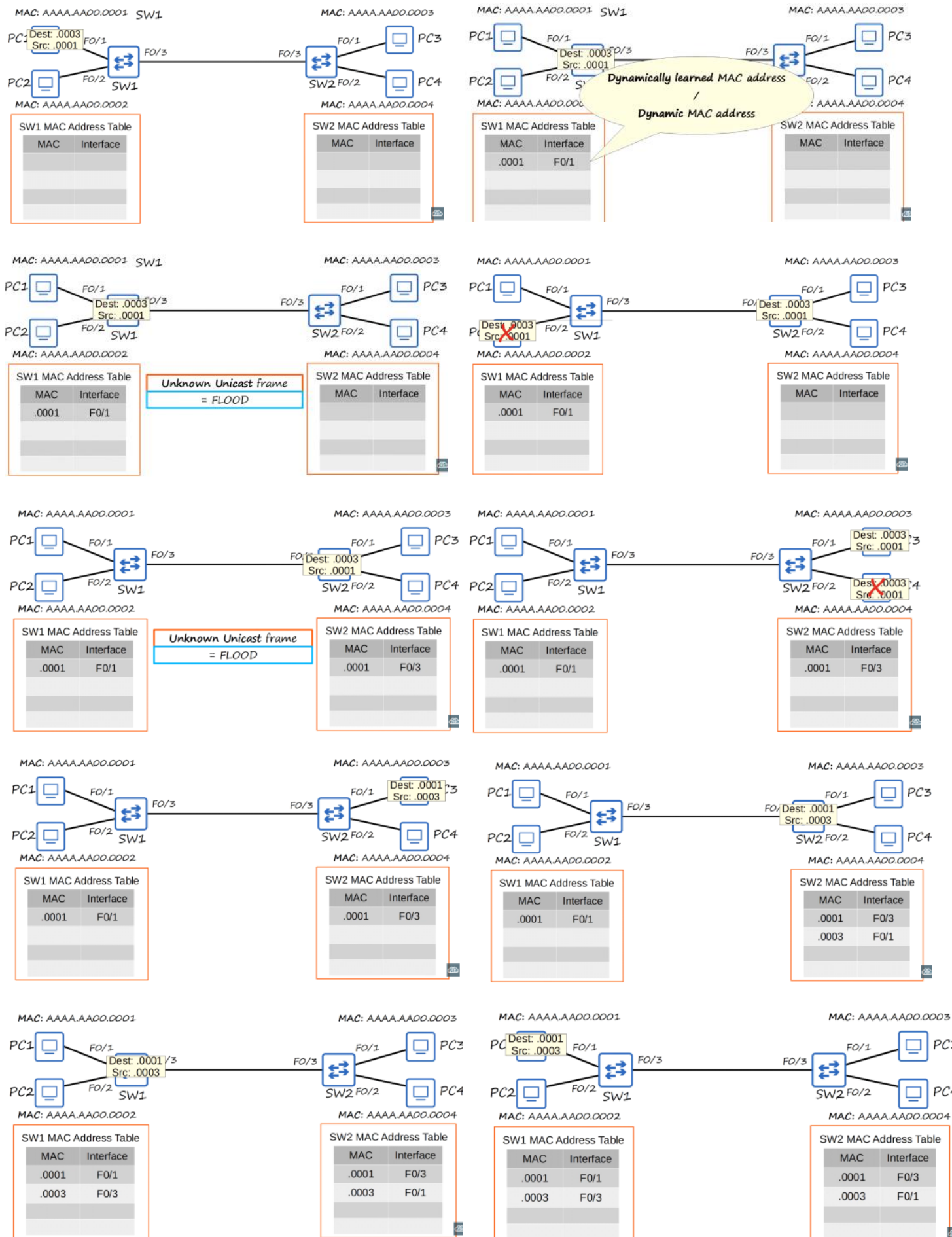
A known unicast frame is an Ethernet frame that is sent from one device to one specific destination device, and the switch already knows which port the destination device is connected to.





# MAC learning & Frame forwarding

19 February 2026 00:49



## MCQs

19 February 2026 01:32

Which field of an Ethernet frame provides receiver clock synchronization?

- a) Preamble
- b) SFD
- c) Type
- d) FCS

How long is the physical address of a network device?

- a) 32 bytes
- b) 32 bits
- c) 48 bytes
- d) 48 bits

What is the OUI of this MAC address? **E8BA.7011.2874**

- a) E8BA
- b) E8BA.70
- c) 7011
- d) E8BA.7011

Which field of an Ethernet frame does a switch use to populate its MAC address table?

- a) Preamble
- b) Length
- c) Source MAC Address
- d) Destination MAC Address

What kind of frame does a switch flood out of all interfaces except the one it was received on?

- a) Unknown unicast
- b) Known unicast
- c) Allcast

- A) Preamble
- D) 48 bits
- B) E8BA.70
- C) Source MAC Address
- A) Unknown unicast